deadline 23 Oct.

- **Prompts:** the competition scores only the numbers. The starter kit's prompt is just a baseline, and any prompt or tool chain is allowed.

Core idea

The QuantiPhy paper shows that VLMs answer from memorised world knowledge: when the known value is scaled to something unrealistic, their accuracy drops 70–80%. A 4B model given the video scores 37 on validation, only 1.3 points above a guesser that never looks at the video.

Most answers are really **(pixel ratio) × (known value)**, sometimes divided by time. So the VLM only does what it is good at (reading text, finding objects), and code does the measuring. The answers follow the known value by construction, which is the property the benchmark paper says VLMs lack.

> One-line version: we don't ask the AI for the answer. We ask it which object is which, track the objects, and compute the answer with a pixel ruler set by the known value.

The pipeline





Every stage caches its output per video, so after a change only the affected stages rerun. A full test run takes about 10 hours on the RTX 2060 and resumes after interruptions (it survived two power cuts).

1 · Parse
rules + Qwen3-VL-4B (text only)

- **Known value by fixed rules**: "speed of **the shark** = 3 m/s" → object shark, quantity speed, value, unit, time ("t=0.6, ball acceleration" → at 0.6 s). Parsed once per video, so every question in a clip shares one reference. The VLM is asked only when no object is named ("gravity acc = 9.8").
- **Question by Qwen** with invented few-shot examples, returning JSON: target object(s), quantity (size / speed / acceleration / distance / displacement / path), time window. Units come from a regex, and keywords override the VLM ("acceleration" in the text wins).
- **Question-meaning rules**: "horizontal/vertical speed" uses only that axis (2D), and "wingspan" uses the long axis.

2 · Ground
Qwen3-VL-4B, 1,024 px frame

- Qwen draws a box for each object on its key frame (the time the question asks about). If the object isn't found, it retries at the quarter points and then on a 12-frame sweep, which catches objects visible only briefly.
- **Telling similar objects apart** ("big bird" / "small bird"): ask for every instance, then choose by size, left/right, top/bottom, depth words (far = smaller), **colour** (HSV pixel rules per colour word), or elimination. If it's still ambiguous, the item goes to the fallback rather than sharing one mask.

3 · Track
SAM 2.1-small, fp16

The boxes seed SAM 2, which propagates pixel masks forward and backward through the clip. Frames and tracking state stay in CPU RAM (otherwise VRAM spills and runs 4–10× slower), and long clips are capped at 96 frames. From each mask per frame: area, centre, bounding box, principal axes, equivalent diameter, and whether it touches the border.

4 · Refix + anchors
frame differencing, Qwen, SAM

- **Motion-guided re-grounding**: a known speed or acceleration means the reference must move. If its track is static (the wrong instance was boxed), find the regions that change between frames and ask Qwen for the object inside them.
- **Distance anchors (3D)**: objects that have a stated distance but are never asked about get grounded and tracked too, so other objects can borrow their distance.

5 · Depth (3D)
MoGe-2 ViT-L

- MoGe's **absolute** depth is unreliable on this data (12–17% of distances within 25%), but its **ratios** are good. An object without a stated distance gets `d_known × MoGe(obj) / MoGe(known)` : 84% within 25% (median error 5%). The transfer is dropped if the known objects disagree by more than 2×, and only one anchor is kept so MoGe noise can't invent motion.
- MoGe also estimates the camera's focal length. Its bias is corrected per type of footage (label-free, see below).

6 · Solve
NumPy / SciPy, CPU

- **2D (pixel ruler):** metres per pixel = known value ÷ the reference's pixel measurement, which can be a size, a speed, or an acceleration. Average speed uses the straight-line distance when the path is nearly straight (mask jitter inflates path length) and the smoothed path otherwise (orbits). Diameter uses the mask's short axis, which motion blur doesn't stretch.
- **Accelerations:** fit over the longest clean **constant-acceleration phase** (rest → fall → bounce is split into phases), using a robust parabola fit whose tolerance adapts to the track's own noise. Precision is checked through the standard error.
- **3D (pinhole camera):** the stated distances are Euclidean, so they're converted to depth along the optical axis per pixel. The one unknown, the focal length *f*, is root-solved per clip so the reference reproduces the known value, and every target is then measured in metres. If the known value can't fix *f*, or gives an impossible field of view (over 118°), or a guard rejects the clip, the corrected MoGe focal length is used instead.

7 · Guards
physics plausibility

- A person's height outside 0.8–2.4 m or a vehicle length outside its class range means the clip's scale is broken, and the whole clip is rejected. Implausible vehicle widths or heights reject only that item.
- Free-fall bound: under a gravity prior, the falling object can't exceed $g \cdot (t+2\ \text{s})$. This caught "189 m/s" from slow-motion clips.
- Regions (floor, lane, road, tile) aren't segmentable objects and go to the fallback. A motion question whose target never moves means the wrong object was boxed. References moving less than 20 px can't set the scale.

8 · Fallback
Qwen3-VL-4B, 16 frames at 448 px

Items the pipeline can't measure get Qwen's direct answer with the official prompt. Unit mismatches are converted ("1.58 m" for a cm question), and a bug in the official answer parser is fixed: every "X m/s^2" reply used to parse as 2.0. Last resort: the median answer per unit (about 46 items). This counts as using validation labels, so it should be removed. Current answer mix on test: **81% measured**, 17% VLM, 1.4% median.

Prompts

Official starter-kit prompt (our fallback)

```
SYSTEM: You are an expert video analyst specializing in physics measurements.
Analyze the video frames carefully and provide ONLY the numerical answer with
units. No explanation or reasoning needed. Format your response as: [value] [unit].
Example: 2.5 cm. Be as accurate as possible ... give me an estimated answer even if
you are not sure.
USER: [video] Given that {prior}. Additionally, you have the following information
about the distance between the objects in the video and the shooting camera:
{depth_info (3D only)} {question}
Please answer the question with numbers and units ONLY. No explanation needed.
```

Our question parser (JSON, invented few-shot examples)

```
You read a physics question about a short video and output JSON only. Keys:
"targets": the object(s) asked about, as short singular phrases. Two objects only
```

```
for a distance between two objects; for "the two X" repeat the phrase twice.
"target_quantity": one of
["length","width","height","diameter","radius","thickness","size","speed","acceleratic
"target_time": [start, end] in seconds; [t, t] for an instant; -1 = end of video;
null for static sizes and whole-video averages.
Example: "What is the velocity of the frisbee at 0.8s in m/s?" → {"targets":
["frisbee"], "target_quantity": "speed", "target_time": [0.8, 0.8]}
```

Grounding (Qwen3-VL native box output, 0–1000 coordinates)

```
Locate the {object} in the image. Output its bounding box as JSON: [{"bbox_2d":
[x1, y1, x2, y2], "label": "{object}"}]. If it is not visible, output [].
(when telling instances apart)  Locate every {head noun} in the image. Output one
bounding box per {head noun} as a JSON list ...
```

Prompts we tested and dropped

```
Mask verifier: [frame with red box] [close-up] Which statement is true? A) the box
tightly contains the whole {obj} B) a different object C) only a part of the {obj}
D) the {obj} plus a lot of other things. Answer with one letter.
Ratio fallback: Question: "{question}". Do not answer it directly. Instead, tell
how many times the {size} of the {reference} the requested quantity is.   (answer =
ratio × known value)
```

What is new or unusual here

1. **Measurement-first with a 4B model.** The VLM never produces measured numbers. Answers scale with the known value by design (the benchmark's central criticism of VLMs).
2. **Deterministic known-value parsing**: fixed rules on the text, not the LLM. This removed a whole class of wrong-reference errors (shark vs boat).
3. **Phase-based acceleration**: a robust, noise-adaptive parabola fit over the clean constant-acceleration phase. It helped D2 most on the push from 64 to 68.
4. **Per-clip focal-length calibration from the known value** in a pinhole model with the stated Euclidean distances, so 3D needs no metric depth network.
5. **MoGe-2 used for ratios only**: relative-depth transfer of stated distances (84% within 25%), plus MoGe's focal length as a fallback camera. This took D3 from 52.2 to 57.5, the biggest single jump.

6. **Label-free self-calibration**: MoGe's focal-length bias per type of footage (×0.96 lab, ×1.2 background-removed, ×0.6 simulation) measured against clips with a reliable size prior. No answers are used.
7. **Label-free error detection**: the same question on variants of one scene (lab original / background-removed / x, simulation / segmented) should get one answer. This exposed unstable 3D calibration (47% disagreement in D3) without test labels.
8. **Motion-guided re-grounding** and **colour / depth-word / elimination rules** to fix wrong-instance grounding without a bigger model.
9. **Error decomposition**: only about 8% of the measurement error is clip-level scale. Most is per-item: which object, which part, which dimension. That steers where effort goes.
10. **Process**: every submission is audited by an independent reviewer agent, which caught real bugs, including 1-based template IDs, "m/s^2" parsed as 2.0, and wrong 3D guard logic.

Leaderboard history (hidden test set, MRA ×100)

VERSION	MAIN CHANGE	AVG	S2	D2	S3	D3
v4	first clean full run: grounding + SAM 2 + 2D/3D solver + guards	55.5	59.3	61.5	51.7	49.7
v6.1	MoGe relative-depth transfer, region and vehicle guards, fuzzy name matching	58.9	62.9	64.2	56.7	52.1
v7	phase-based acceleration, colour/elimination grounding, motion-guided refix	60.3	63.9	67.8	57.2	52.2
v8	MoGe focal length when calibration fails	61.6	63.9	67.8	57.3	57.5
v9	compound names (basketball ↔ ball), track stated-distance objects	61.8	63.9	67.8	58.1	57.4

v10	3D target accelerations over the clean phase	61.8	—	—	—	57.5
v11	diameter = mask short axis	62.0	64.0	68.0	58.2	57.6
v12	MoGe focal bias correction	61.9				
paper	human baseline / GPT-5.1 / Qwen3-VL-32B	55.6 / 53.1 / 46.0				

v13 (question-meaning rules, validation 62.6 → 63.4) is built but not uploaded. Gains have shrunk to tenths per submission because validation (159 items) is too small to tell a +1 fix from a −1 fix before uploading.

Where the points are lost (validation)

Per-item measurement (object, part, dimension, mask)	~55%
VLM fallback items (23% of items, MRA 0.41)	37%
Clip-level scale / camera calibration	~8%

- With a perfect scale in every clip, measured items would only go from 0.69 to 0.74, so calibration work hit diminishing returns.
- Plain sizes (48% of test) and plain speeds (26%) carry most of the loss, as scatter rather than a fixable bias. Exceptions: tiny objects under 20 px come out about 1.46× too big (mask edges and blur), and 3D speeds toward the camera run about 10% low.
- Special question families (horizontal component, gaps between surfaces, wingspans, parts, regions, paths) fail badly but make up only about 5% of test.

Tried and dropped (negative results are part of the story)